Raphaël Tinarrage

Curriculum vitae

Institute of Science and Technology Austria
Central Building, 2nd floor, I01.O2.055
Am Campus 1, Klosterneuburg, Austria
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raphaeltinarrage.github.io

Links: ORCID — ResearchGate — Google Scholar — arXiv — HAL — theses.fr — Lattes — GitHub — YouTube

Academic positions

2024–present **Postdoc**, Institute of Science and Technology Austria (ISTA), Klosterneuburg
Classifying spaces in Topological Data Analysis, in Uli Wagner's group

2021–2024 **Postdoc**, Fundação Getulio Vargas – Escola de Matemática Aplicada (FGV EMAp), Rio de Janeiro
Theory and applications of Topological Data Analysis, supervised by César Camacho

Education

2017–2020 **PhD**, Inria Saclay and Laboratoire de Mathématiques d'Orsay
Topological inference from measures and vector bundles — HAL:tel-02970491
Supervised by Frédéric Chazal and Marc Glisse
Manuscript — slides — reports

2014–2017 **Master's degree**, Université Paris-Saclay
M2 research – Mathematics for life sciences
M1 – Fundamental and applied mathematics & Magistère de mathématiques 2nd year

2015–2016 **Master's degree**, École Normale Supérieure Paris-Saclay
M2 FESUP – Preparation for the *agrégation* examination (teaching diploma)

2013–2014 **Bachelor's degree**, Université Paris-Saclay
L3 – Fundamental and applied mathematics & Magistère de mathématiques 1st year

2011–2013 **Classes préparatoires**, Lycée Camille Pissarro, Pontoise
MPSI & MP

Examinations, competitions, fellowships

09/2024 **ISTA-Fellow: Postdoctoral Program**, Institute of Science and Technology Austria (ISTA)
Two-year fellowship

11/2023 **Competition for Professor Adjunto**, Universidade do Estado do Rio de Janeiro (UERJ)
Rank: 1st (out of 10 participants)

07/2016 **Agrégation externe de mathématiques**, French teaching diploma
Rank: 68th (out of 1896 participants)

Teaching

2024 **Vector Calculus**, FGV EMAp, Rio de Janeiro
2nd year undergraduate course (30 hours)
Course webpage — notes (original document, 180 pages, in Portuguese)

2023 **General and Combinatorial Topology**, FGV EMAp, Rio de Janeiro
Summer course for undergraduate and master's students (26 hours)
Course webpage — notes (original document, 95 pages, in English)

2021 **Topological Data Analysis with Persistent Homology**, FGV EMAp, Rio de Janeiro
Summer course for undergraduate and master's students (22 hours)
Course webpage — notes (original document, 97 pages, in English) — videos

2017–2020 **Statistical Interpretation of Data**, *UE M331, L3 MINT*, Université Paris-Saclay, Orsay
Teaching fellow for undergraduate students

2017–2020 **Modeling Project**, *UE M326, L3 MINT*, Université Paris-Saclay, Orsay
Teaching fellow for undergraduate students

2017–2019 **Ordinary Differential Equations**, *UE M257, L2 BC*, Université Paris-Saclay, Orsay
Teaching fellow for undergraduate students
Course webpage

Mentoring

2022–2023 **Fine-tuning legal language models via annotations**, FGV EMAp, Rio de Janeiro
BSc students: Livia Cales, Victoria Cury, Samanta Duarte, Clara Lopes, Eduardo Portol, João Meirelles, Ana Rosenburg, and Helena Torres

2021–2023 **Data analysis of symmetries with Lie groups**, FGV EMAp, Rio de Janeiro
MSc student: Henrique Ennes

2021–2023 **Machine learning for Súmulas Vinculantes**, FGV EMAp, Rio de Janeiro
BSc students: Beatriz Sabdin Chagas, Carla Marcondes Damian, Ana Clara Macedo Jaccoud, and Pedro Burlini de Oliveira

Academic visits

11/2025 **Anthea Monod Group**, Imperial College, London
Duration: 3 days.

10/2022 **Grupo de Bases de Dados e de Imagens (GBdI)**, Instituto de Ciências Matemáticas e de Computação (ICMC), Universidade de São Paulo (USP), São Carlos
Visualization of temporal graphs with zigzag persistent homology
Host: Agma Juci Machado Traina. Duration: 1 week.

Academic service

2026–present **Organizer** of the ABCDEFG seminar, ISTA

2025–present **Organizer** of the Combinatorics, geometry and topology seminar (CGT), ISTA — webpage

2018–2025 **Reviewer** for Symposium on Computational Geometry (SoCG)

2025 **Reviewer** for Journal of Mathematical Imaging and Vision

2023 **Discussant** at the workshop Transforming the Role of International Courts and Tribunals in a New Era of Adjudication, *FGV Jean Monnet Centre of Excellence*, FGV, Rio de Janeiro — webpage

2023 **Reviewer** for Foundations for Undergraduate Research in Mathematics (FURM)

2023 **Reviewer** for SIAM Journal on Applied Algebra and Geometry (SIAGA)

2022 **Reviewer** for MathSciNet (American Mathematical Society)

2017–2020 **Organizer** of the Workshop MATh.en.JEANS, Collège Alain Fournier, Orsay
Mathematics outreach in college — webpage

Journal articles

05/2026 **Train-Free Segmentation in MRI with Cubical Persistent Homology**
Anton François, **Raphaël Tinarrage**
Journal of Mathematical Imaging and Vision **68**, 20 (2026)
DOI:10.1007/s10851-026-01300-1 — arXiv:2401.01160
22 double column pages, in English

09/2025 **LieDetect: Detection of representation orbits of compact Lie groups from point clouds**
Henrique Ennes, **Raphaël Tinarrage**
Foundations of Computational Mathematics (2025)
DOI:10.1007/s10208-025-09728-4 — arXiv:2309.03086
136 pages, in English

05/2025 **Empirical analysis of binding precedent efficiency in Brazilian Supreme Court via case classification**
Raphaël Tinarrage, Henrique Ennes, Lucas Resck, Lucas T. Gomes, Jean R. Ponciano, Jorge Poco
Artificial Intelligence and Law (2025)
DOI:10.1007/s10506-025-09458-6 — arXiv:2407.07004
67 pages, in English

01/2025 **ZigzagNetVis: Suggesting temporal resolutions for graph visualization using zigzag persistence**
Raphaël Tinarrage, Jean Ponciano, Cláudio Linhares, Agma Traina, Jorge Poco
IEEE Transactions on Visualization and Computer Graphics **31**(10), 6852–6869 (2025)
DOI:10.1109/TVCG.2025.3528197 — arXiv:2304.03828
18+7 (supplementary) double column pages, in English

02/2023 **Recovering the homology of immersed manifolds**
Raphaël Tinarrage
Discrete & Computational Geometry **69**, 659–744 (2023)
DOI:10.1007/s00454-022-00409-5 — arXiv:1912.03033
86 pages, in English

11/2021 **Computing persistent Stiefel-Whitney classes of line bundles**
Raphaël Tinarrage
Journal of Applied and Computational Topology **6**, 65–125 (2022)
DOI:10.1007/s41468-021-00080-4 — arXiv:2005.12543
61 pages, in English

Conference articles

06/2026 **Simplicial approximation to CW complexes with spherical Delaunay triangulations**
Raphaël Tinarrage
International Symposium on Computational Geometry (SoCG 2026)
DOI:10.4230/LIPIcs.SoCG.2026.93 — arXiv:2112.07573
53 pages, in English

09/2022 **O impacto da Súmula Vinculante 26 na diminuição de demanda similar no STF: uma análise quantitativa por modelos de ML**
Beatriz Sabdin Chagas, Carla Marcondes Damian, **Raphaël Tinarrage**
XI Encontro Internacional do CONPEDI Chile, 82–103 (2022)
ISBN:978-65-5648-559-1
22 pages, in Portuguese

09/2022 **Progressão de regime em crimes hediondos no Supremo Tribunal Federal: uma análise empírica pela Súmula Vinculante 26**
Ana Clara Macedo Jaccoud, Pedro Burlini de Oliveira, **Raphaël Tinarrage**
XI Encontro Internacional do CONPEDI Chile, 399–418 (2022)
ISBN:978-65-5648-569-0
20 pages, in Portuguese

06/2020 **DTM-based filtrations**
Hirokazu Anai, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hiroya Inakoshi, **Raphaël Tinarrage**, Yuhei Umeda
International Symposium on Computational Geometry (SoCG 2019) — DOI:10.4230/LIPIcs.SoCG.2019.58
Abel Symposia 15, Springer, Cham. (2020) — DOI:10.1007/978-3-030-43408-3_2
arXiv:1811.04757
33 pages, in English

Posters

06/2022 **Simplicial approximation to CW-complexes in practice**, ATMCS, University of Oxford
Poster

06/2018 **DTM-filtrations**, Algebraic Topology: Methods, Computation and Science, IST Austria
Poster

Selected talks

11/2025 **London-Oxford-Paris TDA Seminar**, City St George's
Simplicial approximation, Delaunay triangulations, and the list homomorphism problem
Slides — program

06/2025 **Applied Algebraic Topology Research Network**, Online
Detection of representation orbits of compact Lie groups from point clouds
Slides — video

02/2025 **Infinite-dimensional Geometry Conference**, Erwin Schrödinger Institute (ESI)
Train-Free Segmentation in MRI with Cubical PH
Slides

07/2024 **XXIII Encontro Brasileiro De Topologia**, Universidade Federal da Bahia (UFBA)
Classifying spaces in TDA
Slides — program

01/2023 **Summer School on Data Science**, FGV EMAp
I. From Topology to Data Analysis: Slides — video
II. Homological inference: Slides — video
III. Persistent Homology: Slides — video

01/2023 **Workshop On Legal Digital Intelligence**, FGV EMAp
TDA and Súmulas Vinculantes
Slides

11/2022 **Seminário ICMC**, Universidade de São Paulo (USP), São Carlos
TDA para escolha de resolução temporal na visualização de grafos
Slides

12/2020 **Applied Algebraic Topology Research Network**, Online
Persistent Stiefel-Whitney classes
Slides — video